EE1003 Introduction to Computer I

Programming Assignment 3: Big Number Calculator

Due Date: 2021/12/05 (Sun.) 23:59

Background

In C++ language, we know that a **`long long int`** variable only use 64 bits (4bytes) to store data, and the range it can represent is $[-2^{63},2^{63}-1]$. Even using **`unsigned long long int`** can only represents the maximum value of $2^{64}- 1$. How do we present a number whose value is greater than $2^{64}– 1$? The answer is Big Number. This homework asks you to write a program to compute three operations (addition, subtraction and multiplication) between two big numbers. The maximum number of digits for each big number is 100 (this means you need a 200-digit big number for the product of a multiplication operation). This is a good time to practice array/vector/string programming skills because we all know how to compute the addition of two numbers by hands. In this assignment, we just want you to do the computation using a programming language C++.

Description

In this programming assignment, you have to implement a calculator which you can choose the functions: 0 for exit, 1 for addition, 2 for subtraction, and 3 for multiplication with big numbers up to 100 digits. At the beginning, the program will show the prompt "1.Addition 2.Subtraction 3.Multiplication 0.Exit". If the user inputs a wrong number, the program will show "Error! Please try again". After the user chooses the function, the program will show the prompt to guide the user to enter two big numbers. Then the program will **output the corresponding result. After that, the program will output the arithmetic mean of the two big numbers.** Arithmetic mean is the average of the two big numbers. The calculator will repeat until the user inputs a 0 at main menu. The program will show "Good bye" and end the program. The calculator must follow the four arithmetic rules. No parentheses will appear at input. In your program, **you must use data structure array, vector or string to store all the numbers.**

Input

1. An integer input as function selection of arithmetic operations: 0 for exit, 1 for addition, 2 for subtraction, and 3 for multiplication
2. Two positive integer sequence for calculating with at most 100 digits.

Output

The program will show a welcome message and prompting message at the beginning. Next, it will show the prompt to let user enter the function selection. After that, the program will show the prompt to let user enter the two big numbers for calculation. Then, the program will show the exact result and arithmetic mean after calculation. The average of the two numbers should be represented as an integer number, all the decimal portion are unconditionally discard. The programmer needs to provide foolproof for function selection. When the program obtains 0 as the input for function, the program will show "Good bye!" and end its execution.

Sample

Note: *Italic* shows the input from keyboard.

```
----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
1
----Now for Addition----
Please enter the two numbers.
11112222333344445555
55554444333322221111
Result: 66666666666666666666
Mean: 33333333333333333333

----Welcome to big number calculator----
Please choose the function
```

```
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
5
Error! Please try again.

----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
3
----Now for Multiplication----
Please enter the two numbers.
111111111
111111111
Result: 12345678987654321
Mean: 111111111

----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
0
Good bye!
```

Bonus

Can your program handle the out-of-range including negative integers or wrong inputs?(2%) Besides, the four arithmetic rules include “division”. Please add the fourth function: division. The two input numbers are still positive, and the output is quotient(6%). If your program can present the remainder, you will get the extra points(2%).

Submission Requirement

You have to submit a source code file (not the entire project) named as StudID_PA3.cpp (ex: 9862534_PA3.cpp) and a report named StudID_Name_PA3_report.pdf (ex: 9862534_ 陳 聿 廣 _PA3_report.pdf). If you implement a bonus version, please separate it from the original version. Name your source code file of the bonus version as StudID_PA3_bonus.cpp (ex: 9862534_PA3_bonus.cpp) and upload to ee-class with the original version. Note that the only acceptable report file format is .pdf, no .doc/.docx or other files are acceptable. **BE SURE to follow the naming rule mentioned above. Otherwise, your program will be not graded.**

Grading

The grading is as follows:

(1) Correctness of your code: 50%

(2) Readability of your code: 10%

(3) The report: 10%

(4) Demo session: 30%

(5) Bonus (at most): 10%

Please submit your assignment on time. Otherwise, the penalty rule will apply:

- Within 72hrs delay: 20% off
- More than 3 days: 0 point

Be sure to attend a demo session (the time will be announced later). If you have questions, please E-mail to both me (andyygchen@ee.ncu.edu.tw) and TA 楊修宜 (hyyang920@gmail.com)

Reference

[1] Chapter 2 ~ Chapter 6, Paul Deitel and Harvey Deitel, “C++ how to program late objects version,” 7th edition, Person 2011.